Explainable Vision Transformers for Histopathological Cancer Detection

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Deep Learning Course

Why Vision Transformers?

Clinical Challenge

- **Cancer misdiagnosis** — common medical error
- Pathologists are **scarce & fatigued**
- **Early detection** saves lives

Why ViT Over CNNs?

- **Global context** — sees whole image relationships
- **Higher accuracy** on fine-grained datasets
- **Richer attention maps** for XAI

Architecture Comparison

CNN

Local receptive field first • Builds global view gradually • Misses long-range patterns

ViT

Global attention from start • All patches interact directly • Captures tissue relationships

Two Classification Tasks

Colon — Binary Classification
Benign vs Adenocarcinoma

Lung — 3-Class Classification
Benign vs Adenocarcinoma vs Squamous Cell Carcinoma

LC25000 Dataset & Histopathology Fundamentals

LC25000 Dataset

25K
Total Images

5K
Per Class

768²
Resolution

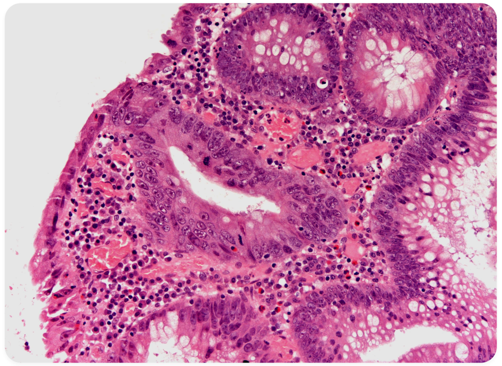
H&E Staining

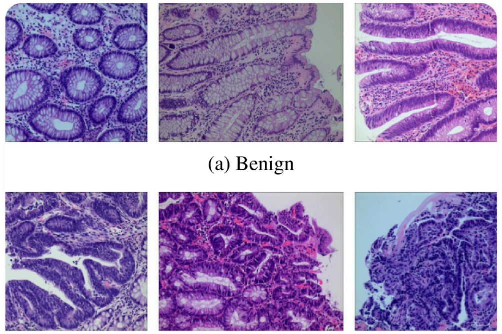
- **Hematoxylin** → Nuclei (Blue/Purple)
- **Eosin** → Cytoplasm (Pink)

Cancer Indicators

Irregular nuclei • Abnormal cell shapes • Atypical tissue architecture

Histopathology Tissue Samples





(a) Benign

(b) Malignant

Benign Tissue

Malignant Tissue

Dataset Organization

Colon
colon_aca • colon_n

Lung
lung_aca • lung_scc • lung_n

ViT-B/16: From Patches to Predictions

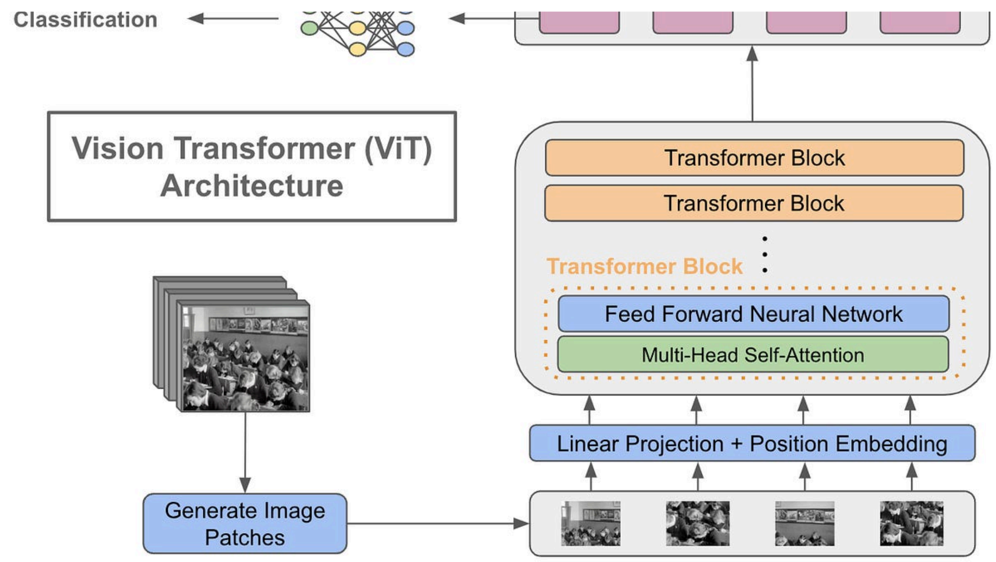
1 **Patch Splitting**
224×224 → 196 patches of 16×16

2 **Patch Embedding**
768 raw values → `768-dim` vector per patch

3 **CLS Token + Positional Encoding**
Prepended token + learnable position embeddings

4 **12 Transformer Blocks**
MHSA (12 heads) + FFN (768→3072→768) + Residual

5 **Classification Head**
Linear(768→2) Colon | Linear(768→3) Lung



Transfer Learning
Pre-trained on **ImageNet-21k**
14M images • 21k categories

Model Size
86M
Parameters

Optimization Pipeline for Medical Imaging

AdamW (Not Adam)

Decoupled weight decay • Proper L2 regularization • Better generalization for Transformers

Cosine Annealing (Not Step Decay)

Smooth decay `2e-4 → 1e-6` • No abrupt drops • Better convergence

Mixed Precision (FP16)

2× faster training • **50% less VRAM** • Same accuracy

Class-Weighted Loss (Lung)

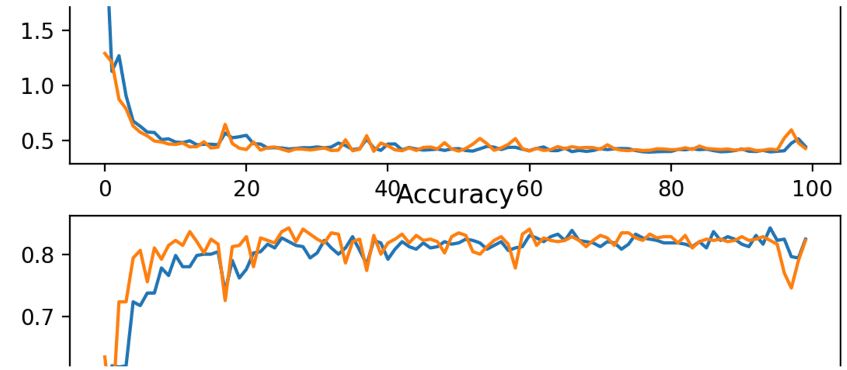
Penalizes errors on harder classes • Compensates for imbalance

Data Augmentation

Horizontal/Vertical Flip
No canonical orientation

Random Rotation (±15°)
Slide placement variation

ColorJitter
Compensates for stain variation between labs



Cosine Annealing: Smooth LR Decay

Clinical-Grade Performance

Colon Classification

Test Accuracy
99.93%

Macro F1
0.9993

Lung Classification

Test Accuracy
99.82%

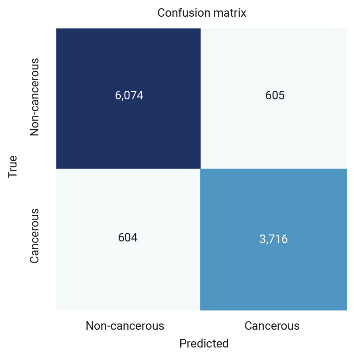
Macro F1
0.9982

! Clinical Priority

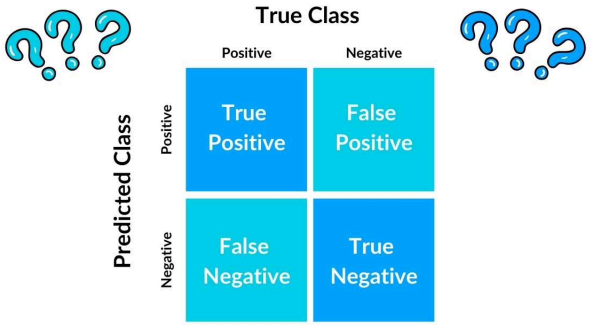
Recall > Precision

Missing cancer (FN) is worse than false alarm (FP)

Colon Confusion Matrix



Lung Confusion Matrix



Macro F1: Why It Matters

Benign	Adenocarcinoma	Squamous Cell
F1 ≈ 0.999	F1 ≈ 0.999	F1 ≈ 0.999

Averages F1 across all classes equally • Most informative for 3-class problems

Transparent AI: From Black Box to Clinical Insight

Attention Rollout

Class-Agnostic

- Propagates attention through `12 layers`
- Shows global attention flow: **WHERE** model looks overall
- Head fusion + discard ratio for noise reduction

Grad-CAM for ViT

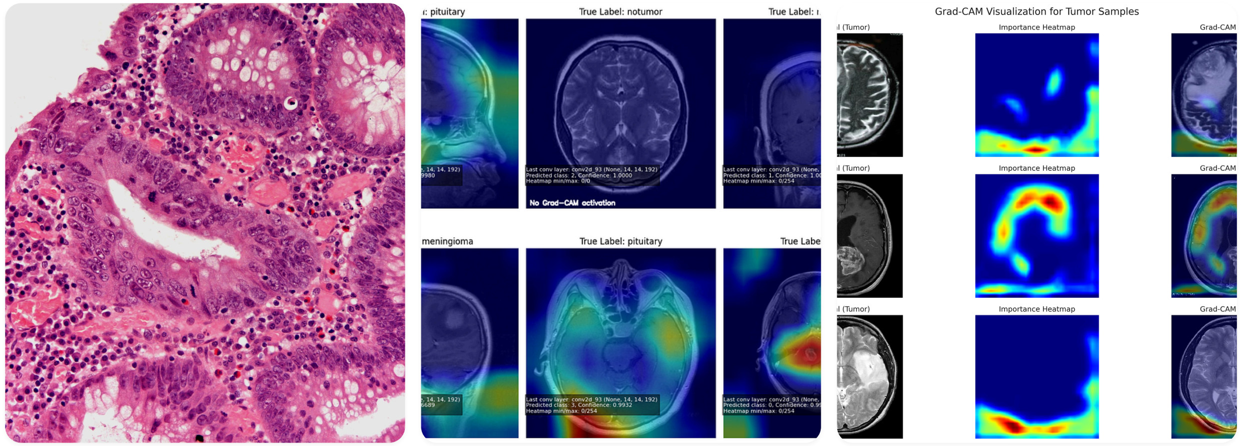
Class-Discriminative

- Gradient-weighted activations from last block
- Shows prediction-specific: **WHAT** drives each prediction
- ReLU keeps only positive contributions

Key Difference

Rollout	Grad-CAM
Same map regardless of class	Different maps per class

XAI Comparison: Original vs Attention Rollout vs Grad-CAM



Original Image
Raw tissue sample

Attention Rollout
Global attention flow

Grad-CAM
Class-specific regions

Clinical Value

Heatmaps reveal model focuses on **histological structures** (not artifacts) • Enables pathologist verification • Builds trust

Bridging Deep Learning and Clinical Practice

Key Contributions

- **99.93%** Colon / **99.82%** Lung
- Novel XAI for ViT in histopathology
- Dual explanation builds clinical trust

Clinical Impact

- Enables pathologist **verification**
- Reduces diagnostic errors & variability
- AI as **second opinion**, not replacement

Multi-Organ Extension

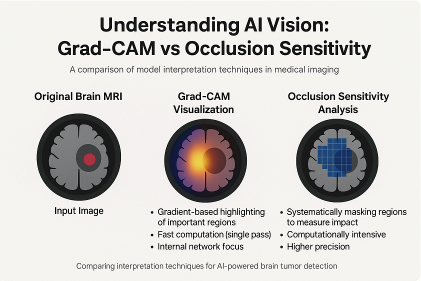
- Organ-agnostic pipeline design
- Auto-generate custom heads
- Consistent XAI across organs

Future: Multi-Modal Fusion Architecture

Histopathology	Radiology	Clinical Data
ViT-B/16	ViT-S/16	MLP Encoder
768-dim	384-dim	128-dim

Fusion: **768 + 384 + 128 = 1280 dim** → MLP → Classification

Clinical metadata: age, smoking history, stage, biomarkers, lymph node involvement



● ViT (Ours) ● CNN Baselines

Multi-Organ Registry

- **Lung:** 3 classes
- **Colon:** 2 classes
- **Breast:** 2 classes
- **Easy extension...**

Thank You

(a) Benign

Questions & Discussion

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